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证明快排实际上有至少 $1 - O(\frac{1}{n})$ 的概率满足 $O(n \log n)$ 的复杂度

$$T(n) = T(q-1) + T(n-q) + O(n)$$

若数组长度为 k , 则 $P(\lfloor \frac{k}{2} \rfloor$ 个元素比任意选取的 i 大) $\leq \frac{1}{2}$

$$\therefore P(\text{不平衡划分}) \leq \frac{1}{2}$$

由于每一层的合并工作量均为 $O(n)$, 设 h 为层数

$$\therefore \text{只需证 } P(h = O(\log n)) \geq 1 - O(\frac{1}{n})$$

$$\text{即证 } P(h < c \log n) = O(\frac{1}{n})$$

$$\text{而 } P(h < c \log n) = P^h(\text{不平衡划分})$$

$$\leq (\frac{1}{2})^{c \log n} = \frac{1}{n^c}$$

$$\text{令 } c=1 \text{ 则有 } P(h < \log n) \leq \frac{1}{n} \Rightarrow P(h < c \log n) = O(\frac{1}{n}) \text{ 成立}$$

$$\therefore P(h = O(\log n)) \geq 1 - O(\frac{1}{n}) \text{ 得证}$$

命题得证

